



FleetRoute — Smart Vehicle Fleet Management System

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Note: Slides (excluding Thank-you and Introduction slide) should not be more than 4.



Problem Statement

The Challenge

Urban transportation networks must dispatch vehicles to passengers across a city while minimising travel distance and managing trip priorities. Manual assignment is slow, error-prone, and does not scale as fleet size and demand grow.

The System Must

- V** Identify the nearest available vehicle to a pickup location
- D** Dispatch it via the shortest path using Dijkstra's Algorithm
- P** Always serve the highest-priority passenger first
- U** Allow the dispatcher to undo (rollback) the last dispatch
- F** Persist all fleet and trip data across sessions



Your Solution

G

City Graph (Adjacency List)

Roads between cities as a weighted graph

Q

Priority Queue (Min-Heap)

Dispatches highest-priority trip first

#

Hash Map

O(1) vehicle lookup by ID

F

File I/O

Persists fleet & trip state (data.txt)

D

Dijkstra + Min-Heap

Finds nearest vehicle & shortest route

U

Stack (Rollback)

Undoes last dispatch, restores state

S

Sort + Binary Search

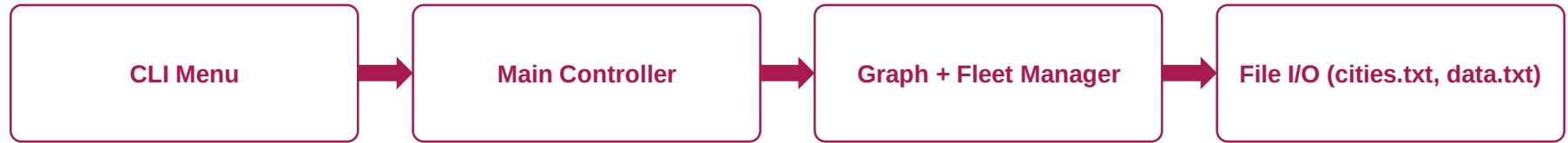
Sorted display & fast ID search

7 DSA Techniques → 1 Dispatch System

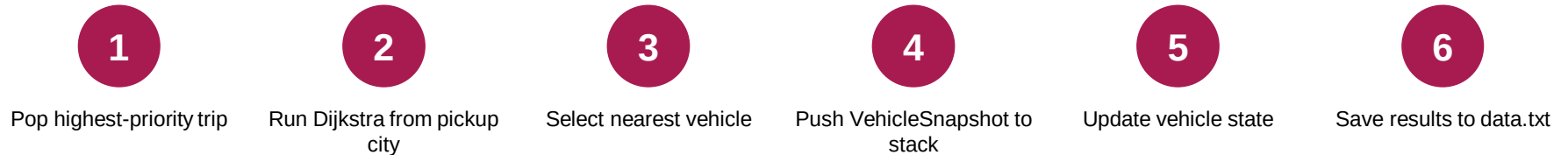


Working (Method)

Architecture



Trip Dispatch Flow



Conclusion

FleetRoute demonstrates how classical DSA concepts translate directly into a practical, real-world system. Every component maps to a course topic:

R

Graphs + Dijkstra

Route optimisation

Q

Priority Queue

Trip scheduling

U

Stack

Undo / rollback

#

Hash Map

Fleet management

S

Sorting + Binary Search

Vehicle lookup & display

Future Scope: Multi-stop routing, real-time traffic weights, GUI front-end





ITM SKILLS
UNIVERSITY

Thank You